Lesson 11: Expectation, Variance, and Joint Distributions

AI/ML Mathematics

Contents

1 Motivation and Overview **2**

2 References **2**

3 Expectation **3**
- 3.1 Expectation for Discrete Random Variables . . . 3
- 3.2 Expectation for Continuous Random Variables . . . 4

4 Variance and Covariance **5**
- 4.1 Variance Properties . . . 5
- 4.2 Covariance . . . 5
- 4.3 Correlation Coefficient . . . 7

5 Joint Distributions **7**
- 5.1 Joint PMF and PDF . . . 7
- 5.2 Marginal Distributions . . . 8
- 5.3 Independence of Random Variables . . . 8
- 5.4 Conditional Distributions . . . 8

6 Conditional Expectation and the Tower Property **8**
- 6.1 Law of Total Variance . . . 10

7 Exercises **10**

8 Summary **11**

9 Further Reading **12**

1 Motivation and Overview

An RL agent estimates the value of a state as an expected sum of future rewards: $V^\pi(s) = \mathbb{E}\big[\sum_{t=0}^{\infty} \gamma^t R_t \mid S_0 = s\big]$. This expression involves an expectation over many random variables (the rewards R_t), decomposes as a sum of individual expectations by linearity, and connects values at successive time steps through the tower property $\mathbb{E}[V^\pi(S_{t+1}) \mid S_t] = \mathbb{E}[\mathbb{E}[\sum \gamma^k R_{t+1+k} \mid S_{t+1}] \mid S_t]$. To make such manipulations rigorous, we need a systematic treatment of expectation, variance, covariance, and the way multiple random variables interact through joint distributions. How does the expectation of a sum relate to the sum of expectations? When does the variance of a sum simplify? How do we condition on one random variable to compute the distribution of another?

The tools developed in this lesson will let us prove the linearity of expectation without assuming independence, establish the variance decomposition for sums, define joint and conditional distributions, and derive the tower property of conditional expectation – the algebraic engine behind the Bellman equation.

Why Expectation and Joint Distributions Matter for RL/ML

- **Value function decomposition**: The Bellman equation $V^\pi(s) = \mathbb{E}[R + \gamma V^\pi(S') \mid S = s]$ uses linearity of expectation to split a value function into an immediate reward and a discounted future value.
- **Bellman equation via the tower property**: The recursive structure of value functions follows from the law of total expectation (tower property), which allows conditioning on the next state.
- **REINFORCE variance reduction**: The baseline subtraction $\nabla J \approx \mathbb{E}[\nabla \log \pi \cdot (G - b)]$ uses the fact that adding a baseline $b(s)$ does not change the expectation (by linearity) but can reduce variance (by the variance of a difference formula).

Prerequisites. Random variables, probability mass functions, probability density functions, cumulative distribution functions, and common distributions (Lesson 10). Conditional probability and independence of events (Lesson 9).

Learning objectives. After completing this lesson, the student will be able to:

- *State* and *prove* the linearity of expectation for both discrete and continuous random variables, without assuming independence.
- *Compute* expectations using the Law of the Unconscious Statistician (LOTUS).
- *Derive* the computational formula $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ and the affine transformation rule $\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X)$.
- *Define* covariance and correlation, and *prove* the variance of a sum formula.
- *Construct* an example showing that uncorrelated random variables need not be independent.
- *Define* joint, marginal, and conditional distributions and *prove* the law of total expectation (tower property) and the law of total variance.

2 References

- Ross [2019, Chapters 4, 6–7] – Expectation, variance, jointly distributed random variables, conditional distributions.

- Blitzstein and Hwang [2019, Chapters 4, 6–7] – Expectation, moments, joint distributions, conditional expectation with many examples from applied probability.
- Grimmett and Stirzaker [2020, Chapter 3] – Discrete and continuous random variables, expectation, variance, generating functions.
- Bertsekas and Tsitsiklis [2008, Chapters 2–4] – Expectation, variance, multiple random variables, conditioning.
- Durrett [2019, Sections 1.5–1.6] – Expectation and conditional expectation in the measure-theoretic framework.

3 Expectation

We develop the properties of expectation for discrete random variables first, then extend to the continuous case.

3.1 Expectation for Discrete Random Variables

Recall from Lesson 10 (Definition 4.2) that if X is a discrete random variable with PMF p_X, then $\mathbb{E}[X] = \sum_x x\, p_X(x)$ whenever the sum converges absolutely.

Theorem 3.1 (Law of the Unconscious Statistician – Discrete Case)**.** *Let X be a discrete random variable and $g\colon \mathbb{R} \to \mathbb{R}$ any function. Then*

$$\mathbb{E}[g(X)] = \sum_x g(x)\, p_X(x),$$

provided the sum converges absolutely.

Proof. Define $Y = g(X)$. The random variable Y takes values in $\{g(x) : x \in \text{range}(X)\}$. For each value y in the range of Y,

$$p_Y(y) = \mathbb{P}(Y = y) = \mathbb{P}(g(X) = y) = \sum_{\{x : g(x) = y\}} p_X(x).$$

Therefore

$$\mathbb{E}[Y] = \sum_y y\, p_Y(y) = \sum_y y \sum_{\{x : g(x) = y\}} p_X(x) = \sum_y \sum_{\{x : g(x) = y\}} g(x)\, p_X(x) = \sum_x g(x)\, p_X(x),$$

where the last equality rearranges the double sum: each x in the range of X appears exactly once, under the unique $y = g(x)$. □

Linearity of Expectation

Linearity of expectation is arguably the single most useful property in probability. It states that $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$ for *any* random variables X and Y (with finite expectations) and constants $a, b \in \mathbb{R}$. Crucially, no independence assumption is needed. This property underlies the decomposition of value functions, the analysis of Monte Carlo estimators, and virtually every expected reward computation in RL.

Theorem 3.2 (Linearity of Expectation – Discrete Case)**.** *Let X and Y be discrete random variables defined on the same probability space with finite expectations. Then for any $a, b \in \mathbb{R}$,*

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

Proof. Let $p_{X,Y}(x,y) = \mathbb{P}(X = x, Y = y)$ be the joint PMF. By the definition of expectation applied to the random variable $aX + bY$,

$$\begin{aligned}\mathbb{E}[aX + bY] &= \sum_x \sum_y (ax + by)\, p_{X,Y}(x,y) \\ &= a \sum_x \sum_y x\, p_{X,Y}(x,y) + b \sum_x \sum_y y\, p_{X,Y}(x,y).\end{aligned}$$

For the first term, note that $\sum_y p_{X,Y}(x,y) = p_X(x)$ (the marginal PMF of X), so

$$\sum_x \sum_y x\, p_{X,Y}(x,y) = \sum_x x \sum_y p_{X,Y}(x,y) = \sum_x x\, p_X(x) = \mathbb{E}[X].$$

Similarly, $\sum_x p_{X,Y}(x,y) = p_Y(y)$, so the second term equals $\mathbb{E}[Y]$. Combining: $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$. □

Remark 3.3. The proof extends immediately to any finite sum: for random variables $X_1, \ldots, X_n$ and constants $a_1, \ldots, a_n$,

$$\mathbb{E}\left[\sum_{i=1}^n a_i X_i\right] = \sum_{i=1}^n a_i \mathbb{E}[X_i].$$

This is proved by induction on n, applying Theorem 3.2 at each step.

Example 3.4 (Indicator Expectation)**.** Let A be an event and $\mathbf{1}_A$ the indicator random variable with $\mathbf{1}_A(\omega) = 1$ if $\omega \in A$ and 0 otherwise. Then $\mathbf{1}_A \sim \text{Bernoulli}(\mathbb{P}(A))$, so $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$. This links probability to expectation: every probability is the expectation of an indicator.

Example 3.5 (Expected Number of Fixed Points)**.** A random permutation σ of $\{1, 2, \ldots, n\}$ is chosen uniformly. Let X be the number of fixed points ($\sigma(i) = i$). Write $X = \sum_{i=1}^n \mathbf{1}_{\{\sigma(i)=i\}}$. Each indicator has expectation $\mathbb{P}(\sigma(i) = i) = 1/n$ (since each element is equally likely to land in any position). By linearity, $\mathbb{E}[X] = n \cdot (1/n) = 1$, regardless of n. Note that the indicators are *not* independent, but linearity of expectation applies nonetheless.

3.2 Expectation for Continuous Random Variables

Theorem 3.6 (LOTUS and Linearity – Continuous Case)**.** *Let X be a continuous random variable with PDF f_X and $g\colon \mathbb{R} \to \mathbb{R}$. Then $\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$, provided the integral converges absolutely. In particular, linearity holds: for jointly continuous X, Y with joint PDF $f_{X,Y}$,*

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

Proof. The LOTUS formula is analogous to the discrete case (Theorem 3.1): one integrates $g(x)f_X(x)$ instead of summing $g(x)p_X(x)$.

For linearity,

$$\begin{aligned}\mathbb{E}[aX + bY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (ax + by) f_{X,Y}(x,y)\, dx\, dy \\ &= a \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x\, f_{X,Y}(x,y)\, dx\, dy + b \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y\, f_{X,Y}(x,y)\, dx\, dy \\ &= a \int_{-\infty}^{\infty} x\, f_X(x)\, dx + b \int_{-\infty}^{\infty} y\, f_Y(y)\, dy = a\mathbb{E}[X] + b\mathbb{E}[Y],\end{aligned}$$

where $f_X(x) = \int f_{X,Y}(x,y)\, dy$ is the marginal PDF of X. □

Linearity and Value Function Decomposition

The Bellman equation for a policy π in a discounted MDP is

$$V^\pi(s) = \mathbb{E}\big[R_t + \gamma V^\pi(S_{t+1}) \mid S_t = s\big] = \mathbb{E}[R_t \mid S_t = s] + \gamma\,\mathbb{E}[V^\pi(S_{t+1}) \mid S_t = s].$$

The second equality is a direct application of linearity: the expectation of the sum (immediate reward plus discounted future value) equals the sum of expectations. No independence between R_t and S_{t+1} is assumed – they may be highly correlated (e.g., higher rewards might occur in states with better future prospects).

We now turn to variance and its properties.

4 Variance and Covariance

4.1 Variance Properties

Recall from Lesson 10 that $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$ and the computational formula $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ (Lesson 10, Proposition 4.4).

Proposition 4.1 (Affine Transformation of Variance)**.** *For any random variable X with finite variance and constants $a, b \in \mathbb{R}$,*

$$\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X).$$

In particular, adding a constant does not change the variance, and scaling by a scales the variance by a^2.

Proof. Since $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$ by linearity,

$$\mathrm{Var}(aX + b) = \mathbb{E}\big[(aX + b - a\mathbb{E}[X] - b)^2\big] = \mathbb{E}\big[a^2(X - \mathbb{E}[X])^2\big] = a^2\mathbb{E}\big[(X - \mathbb{E}[X])^2\big] = a^2\,\mathrm{Var}(X).$$

Example 4.2 (Standardization)**.** If $\mathbb{E}[X] = \mu$ and $\mathrm{Var}(X) = \sigma^2 > 0$, define the *standardized* variable $Z = (X - \mu)/\sigma$. Then $\mathbb{E}[Z] = 0$ and $\mathrm{Var}(Z) = \mathrm{Var}(X)/\sigma^2 = 1$.

4.2 Covariance

Definition 4.3 (Covariance)**.** The *covariance* of random variables X and Y (with finite second moments) is

$$\mathrm{Cov}(X, Y) = \mathbb{E}\big[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\big].$$

Proposition 4.4 (Computational Formula for Covariance)**.**

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Proof. Let $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$. Then

$$\begin{aligned}\mathrm{Cov}(X, Y) &= \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY - \mu_X Y - X\mu_Y + \mu_X\mu_Y] \\ &= \mathbb{E}[XY] - \mu_X\mathbb{E}[Y] - \mathbb{E}[X]\mu_Y + \mu_X\mu_Y = \mathbb{E}[XY] - \mu_X\mu_Y.\end{aligned}$$

□

Proposition 4.5 (Properties of Covariance)**.** *For random variables X, Y, Z with finite second moments and constants $a, b \in \mathbb{R}$:*

(i) ***Symmetry:*** $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$.
(ii) ***Self-covariance:*** $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$.
(iii) ***Bilinearity:*** $\mathrm{Cov}(aX + bY, Z) = a\,\mathrm{Cov}(X, Z) + b\,\mathrm{Cov}(Y, Z)$.

(iv) ***Constant:*** $\mathrm{Cov}(X, c) = 0$ *for any constant* c.

Proof. **(i)** By commutativity of multiplication: $(X - \mathbb{E}[X])(Y - \mathbb{E}[Y]) = (Y - \mathbb{E}[Y])(X - \mathbb{E}[X])$.
(ii) $\mathrm{Cov}(X, X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathrm{Var}(X)$.
(iii) Using the computational formula:

$$\begin{aligned}\mathrm{Cov}(aX + bY, Z) &= \mathbb{E}[(aX + bY)Z] - \mathbb{E}[aX + bY]\mathbb{E}[Z] \\ &= a\mathbb{E}[XZ] + b\mathbb{E}[YZ] - (a\mathbb{E}[X] + b\mathbb{E}[Y])\mathbb{E}[Z] \\ &= a(\mathbb{E}[XZ] - \mathbb{E}[X]\mathbb{E}[Z]) + b(\mathbb{E}[YZ] - \mathbb{E}[Y]\mathbb{E}[Z]) \\ &= a\ \mathrm{Cov}(X, Z) + b\ \mathrm{Cov}(Y, Z).\end{aligned}$$

(iv) $\mathrm{Cov}(X, c) = \mathbb{E}[Xc] - \mathbb{E}[X]\mathbb{E}[c] = c\mathbb{E}[X] - c\mathbb{E}[X] = 0$. □

Theorem 4.6 (Variance of a Sum)**.** *For random variables* $X_1, \ldots, X_n$ *with finite second moments,*

$$\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathrm{Var}(X_i) + 2 \sum_{1 \le i < j \le n} \mathrm{Cov}(X_i, X_j).$$

Proof. Let $S = \sum_{i=1}^{n} X_i$ and $\mu_i = \mathbb{E}[X_i]$. Then $\mathbb{E}[S] = \sum_i \mu_i$ and

$$\begin{aligned}\mathrm{Var}(S) &= \mathbb{E}\left[\left(\sum_{i=1}^{n}(X_i - \mu_i)\right)^2\right] = \mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{n}(X_i - \mu_i)(X_j - \mu_j)\right] \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n}\mathbb{E}\big[(X_i - \mu_i)(X_j - \mu_j)\big] = \sum_{i=1}^{n}\sum_{j=1}^{n}\mathrm{Cov}(X_i, X_j).\end{aligned}$$

Separating diagonal terms ($i = j$) from off-diagonal terms ($i \neq j$) and using symmetry $\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$:

$$\mathrm{Var}(S) = \sum_{i=1}^{n} \mathrm{Var}(X_i) + 2 \sum_{1 \le i < j \le n} \mathrm{Cov}(X_i, X_j).$$ □

Corollary 4.7 (Variance of Independent Sum)**.** *If* $X_1, \ldots, X_n$ *are pairwise independent (i.e.,* $\mathrm{Cov}(X_i, X_j) = 0$ *for* $i \neq j$*; see Proposition 4.8 below), then*

$$\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathrm{Var}(X_i).$$

Proposition 4.8 (Independent Implies Uncorrelated)**.** *If* X *and* Y *are independent random variables with finite second moments, then* $\mathrm{Cov}(X, Y) = 0$.

Proof. Independence of X and Y implies $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$. (For discrete random variables: $\mathbb{E}[XY] = \sum_x \sum_y xy\, p_{X,Y}(x, y) = \sum_x \sum_y xy\, p_X(x) p_Y(y) = (\sum_x x\, p_X(x))(\sum_y y\, p_Y(y)) = \mathbb{E}[X]\mathbb{E}[Y]$; the continuous case is analogous with integrals and the factorization $f_{X,Y}(x, y) = f_X(x) f_Y(y)$.) By the computational formula, $\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0$. □

Uncorrelated Does Not Imply Independent

The converse of Proposition 4.8 is false: $\mathrm{Cov}(X, Y) = 0$ does not imply that X and Y are independent. Covariance measures only *linear* association. Two variables can be functionally related yet uncorrelated.

Example 4.9 (Uncorrelated but Dependent)**.** Let $X \sim \mathrm{Uniform}(-1, 1)$ and $Y = X^2$. Then Y is completely determined by X (maximal dependence). However, $\mathbb{E}[X] = 0$, so $\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \mathbb{E}[X^3]$. Since $x^3 f_X(x) = x^3/(2)$ is an odd function on $(-1, 1)$, $\mathbb{E}[X^3] = \int_{-1}^{1} x^3/(2)\, dx = 0$. Thus $\mathrm{Cov}(X, Y) = 0$ even though Y is a deterministic function of X.

4.3 Correlation Coefficient

Definition 4.10 (Correlation Coefficient)**.** The *(Pearson) correlation coefficient* of X and Y (with $\mathrm{Var}(X) > 0$ and $\mathrm{Var}(Y) > 0$) is

$$\rho(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}.$$

Proposition 4.11. *$-1 \le \rho(X, Y) \le 1$, with $|\rho(X, Y)| = 1$ if and only if $Y = aX + b$ almost surely for some constants $a \neq 0, b$.*

Proof. For any $t \in \mathbb{R}$, consider the non-negative quantity

$$0 \le \mathrm{Var}(X + tY) = \mathrm{Var}(X) + 2t\ \mathrm{Cov}(X, Y) + t^2\,\mathrm{Var}(Y).$$

This is a quadratic in t with a non-negative minimum, so its discriminant must be non-positive: $4\,\mathrm{Cov}(X, Y)^2 - 4\,\mathrm{Var}(X)\,\mathrm{Var}(Y) \le 0$, which gives $\mathrm{Cov}(X, Y)^2 \le \mathrm{Var}(X)\,\mathrm{Var}(Y)$ and hence $|\rho(X, Y)| \le 1$.

Equality $|\rho| = 1$ holds if and only if the discriminant is zero, meaning the quadratic $\mathrm{Var}(X + tY)$ touches zero at some $t_0 = -\mathrm{Cov}(X, Y)/\mathrm{Var}(Y)$. Then $\mathrm{Var}(X + t_0Y) = 0$, so $X + t_0Y = c$ almost surely for some constant c, giving $X = -t_0Y + c$, i.e., $Y = aX + b$ with $a = -1/t_0 \neq 0$. □

Variance Reduction in REINFORCE

The REINFORCE gradient estimator is $\hat{g} = \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(A_t \mid S_t) \cdot (G_t - b(S_t))$, where G_t is the return and $b(S_t)$ is a baseline. By linearity,

$$\mathbb{E}[\nabla \log \pi \cdot b(S)] = \mathbb{E}[\nabla \log \pi] \cdot b(S) = 0$$

(under regularity conditions), so the baseline does not bias the estimator. However, by the variance formula, $\mathrm{Var}(\nabla \log \pi \cdot (G - b)) = \mathrm{Var}(\nabla \log \pi \cdot G) - 2\,\mathrm{Cov}(\nabla \log \pi \cdot G, \nabla \log \pi \cdot b) + \mathrm{Var}(\nabla \log \pi \cdot b)$. Choosing b to minimize this variance (often $b(s) \approx V^\pi(s)$) is a direct application of the covariance and variance algebra developed in this section.

5 Joint Distributions

We now formalize the joint behavior of two or more random variables.

5.1 Joint PMF and PDF

Definition 5.1 (Joint PMF)**.** Let X and Y be discrete random variables. The *joint PMF* is

$$p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y).$$

It satisfies $p_{X,Y}(x, y) \ge 0$ for all (x, y) and $\sum_x \sum_y p_{X,Y}(x, y) = 1$.

Definition 5.2 (Joint PDF)**.** Random variables X and Y are *jointly continuous* if there exists a non-negative function $f_{X,Y} \colon \mathbb{R}^2 \to [0, \infty)$ such that for every region $B \subseteq \mathbb{R}^2$,

$$\mathbb{P}((X, Y) \in B) = \iint_B f_{X,Y}(x, y)\,dx\,dy.$$

The function $f_{X,Y}$ is the *joint PDF*, satisfying $\iint_{\mathbb{R}^2} f_{X,Y}(x, y)\,dx\,dy = 1$.

5.2 Marginal Distributions

Definition 5.3 (Marginal Distributions)**.** The *marginal PMF* of X from the joint PMF is $p_X(x) = \sum_y p_{X,Y}(x, y)$. The *marginal PDF* of X from the joint PDF is $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dy$.

Example 5.4 (Joint and Marginal PMF)**.** Let (X, Y) have the joint PMF given by the following table:

$p_{X,Y}$	$Y = 0$	$Y = 1$	p_X
$X = 0$	0.2	0.3	0.5
$X = 1$	0.1	0.4	0.5
p_Y	0.3	0.7	

The marginal PMFs are $p_X(0) = 0.5$, $p_X(1) = 0.5$, $p_Y(0) = 0.3$, $p_Y(1) = 0.7$.

Example 5.5 (Joint Uniform Distribution)**.** Let (X, Y) be uniformly distributed on the unit square $[0, 1]^2$, so $f_{X,Y}(x, y) = 1$ for $(x, y) \in [0, 1]^2$ and 0 otherwise. The marginal of X is $f_X(x) = \int_0^1 1\, dy = 1$ for $x \in [0, 1]$, i.e., $X \sim \text{Uniform}(0, 1)$, and similarly for Y.

5.3 Independence of Random Variables

Definition 5.6 (Independence of Random Variables)**.** Discrete random variables X and Y are *independent* if $p_{X,Y}(x, y) = p_X(x)\, p_Y(y)$ for all x, y. Continuous random variables X and Y are *independent* if $f_{X,Y}(x, y) = f_X(x)\, f_Y(y)$ for all x, y.

Example 5.7 (Testing Independence from a Joint PMF)**.** Continuing Example 5.4, check whether X and Y are independent. We need $p_{X,Y}(x, y) = p_X(x)p_Y(y)$ for all (x, y). At $(0, 0)$: $p_{X,Y}(0, 0) = 0.2$ but $p_X(0)p_Y(0) = 0.5 \times 0.3 = 0.15 \neq 0.2$. Therefore X and Y are *not* independent.

5.4 Conditional Distributions

Definition 5.8 (Conditional PMF)**.** If X and Y are discrete with $p_Y(y) > 0$, the *conditional PMF of X given $Y = y$* is

$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

Definition 5.9 (Conditional PDF)**.** If X and Y are jointly continuous with $f_Y(y) > 0$, the *conditional PDF of X given $Y = y$* is

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

Example 5.10 (Conditional PMF Computation)**.** From Example 5.4, the conditional PMF of X given $Y = 1$ is

$$p_{X|Y}(0 \mid 1) = \frac{0.3}{0.7} = \frac{3}{7}, \qquad p_{X|Y}(1 \mid 1) = \frac{0.4}{0.7} = \frac{4}{7}.$$

Given $Y = 1$, the conditional distribution of X is Bernoulli(4/7).

6 Conditional Expectation and the Tower Property

Definition 6.1 (Conditional Expectation Given an Event)**.** Let X be a random variable and B an event with $\mathbb{P}(B) > 0$. The *conditional expectation of X given B* is

$$\mathbb{E}[X \mid B] = \sum_x x\, \mathbb{P}(X = x \mid B) \quad \text{(discrete)}$$

or $\mathbb{E}[X \mid B] = \int x\, f_{X|B}(x)\, dx$ (continuous).

Definition 6.2 (Conditional Expectation Given a Random Variable)**.** Let X and Y be random variables. The *conditional expectation of X given $Y = y$* is

$$\mathbb{E}[X \mid Y = y] = \begin{cases} \sum_x x\, p_{X|Y}(x \mid y) & \text{(discrete)}, \\ \int_{-\infty}^{\infty} x\, f_{X|Y}(x \mid y)\, dx & \text{(continuous)}. \end{cases}$$

The *conditional expectation of X given Y*, denoted $\mathbb{E}[X \mid Y]$, is the random variable obtained by evaluating $\mathbb{E}[X \mid Y = y]$ at Y: it equals $h(Y)$ where $h(y) = \mathbb{E}[X \mid Y = y]$.

Law of Total Expectation (Tower Property)

The law of total expectation, also called the tower property or iterated expectation, states that $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$. This identity is the algebraic backbone of the Bellman equation: by conditioning on the next state S_{t+1}, the value function at time t decomposes into an immediate reward and a conditional future value.

Theorem 6.3 (Law of Total Expectation)**.** *Let X and Y be random variables with $\mathbb{E}[|X|] < \infty$. Then*

$$\mathbb{E}\big[\mathbb{E}[X \mid Y]\big] = \mathbb{E}[X].$$

Proof. We prove the discrete case; the continuous case is analogous with integrals replacing sums.

Let $h(y) = \mathbb{E}[X \mid Y = y] = \sum_x x\, p_{X|Y}(x \mid y)$. Then $\mathbb{E}[X \mid Y] = h(Y)$ is a function of Y, so

$$\begin{aligned} \mathbb{E}\big[\mathbb{E}[X \mid Y]\big] &= \mathbb{E}[h(Y)] = \sum_y h(y)\, p_Y(y) \\ &= \sum_y \left(\sum_x x\, p_{X|Y}(x \mid y) \right) p_Y(y) \\ &= \sum_y \sum_x x\, \frac{p_{X,Y}(x,y)}{p_Y(y)}\, p_Y(y) \\ &= \sum_y \sum_x x\, p_{X,Y}(x,y) = \sum_x x \sum_y p_{X,Y}(x,y) = \sum_x x\, p_X(x) = \mathbb{E}[X]. \end{aligned}$$

The key step is that $p_{X|Y}(x \mid y) \cdot p_Y(y) = p_{X,Y}(x,y)$ cancels the denominator, recovering the joint PMF. □

Example 6.4 (Tower Property in Action)**.** Roll a fair die (Y), then toss a fair coin Y times. Let X be the number of heads. Given $Y = y$, we have $X \mid (Y = y) \sim \mathrm{Bin}(y, 1/2)$, so $\mathbb{E}[X \mid Y = y] = y/2$, and thus $\mathbb{E}[X \mid Y] = Y/2$. By the tower property, $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[Y/2] = \mathbb{E}[Y]/2 = 3.5/2 = 1.75$.

Tower Property and the Bellman Equation

The Bellman equation for the state-value function is derived by conditioning on the first transition:

$$\begin{aligned} V^\pi(s) &= \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R_t \;\middle|\; S_0 = s \right] = \mathbb{E}\left[R_0 + \gamma \sum_{t=1}^{\infty} \gamma^{t-1} R_t \;\middle|\; S_0 = s \right] \\ &= \mathbb{E}[R_0 \mid S_0 = s] + \gamma\, \mathbb{E}\left[\mathbb{E}\left[\sum_{t=1}^{\infty} \gamma^{t-1} R_t \;\middle|\; S_1 \right] \;\middle|\; S_0 = s \right] \\ &= \mathbb{E}[R_0 \mid S_0 = s] + \gamma\, \mathbb{E}[V^\pi(S_1) \mid S_0 = s]. \end{aligned}$$

The second line applies the tower property: the outer expectation is over S_1 (given $S_0 = s$), and the inner expectation is over the future trajectory given S_1. Without the tower property, the recursive decomposition of value functions would not be justified.

Example 6.5 (Conditional Expectation with Continuous Variables)**.** Let $X \sim \text{Exp}(1)$ and $Y \mid (X = x) \sim \text{Uniform}(0, x)$. Then $\mathbb{E}[Y \mid X = x] = x/2$, so $\mathbb{E}[Y \mid X] = X/2$. By the tower property, $\mathbb{E}[Y] = \mathbb{E}[X/2] = \mathbb{E}[X]/2 = 1/2$.

6.1 Law of Total Variance

Theorem 6.6 (Law of Total Variance)**.** *Let X and Y be random variables with $\mathbb{E}[X^2] < \infty$. Then*

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y]),$$

where $\text{Var}(X \mid Y)$ is the random variable $\mathbb{E}[X^2 \mid Y] - (\mathbb{E}[X \mid Y])^2$.

Proof. By the computational formula for variance, $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$. We compute each term using the tower property.

First term. By the tower property, $\mathbb{E}[X^2] = \mathbb{E}[\mathbb{E}[X^2 \mid Y]]$.

Second term. By the tower property, $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$, so $(\mathbb{E}[X])^2 = (\mathbb{E}[\mathbb{E}[X \mid Y]])^2$.

Now consider the right-hand side. The conditional variance is $\text{Var}(X \mid Y) = \mathbb{E}[X^2 \mid Y] - (\mathbb{E}[X \mid Y])^2$. Taking expectations:

$$\mathbb{E}[\text{Var}(X \mid Y)] = \mathbb{E}[\mathbb{E}[X^2 \mid Y]] - \mathbb{E}[(\mathbb{E}[X \mid Y])^2] = \mathbb{E}[X^2] - \mathbb{E}[(\mathbb{E}[X \mid Y])^2].$$

The variance of the conditional expectation is

$$\text{Var}(\mathbb{E}[X \mid Y]) = \mathbb{E}[(\mathbb{E}[X \mid Y])^2] - (\mathbb{E}[\mathbb{E}[X \mid Y]])^2 = \mathbb{E}[(\mathbb{E}[X \mid Y])^2] - (\mathbb{E}[X])^2.$$

Adding:

$$\begin{aligned}\mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y]) &= \big(\mathbb{E}[X^2] - \mathbb{E}[(\mathbb{E}[X \mid Y])^2]\big) + \big(\mathbb{E}[(\mathbb{E}[X \mid Y])^2] - (\mathbb{E}[X])^2\big) \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \text{Var}(X).\end{aligned}$$

□

Example 6.7 (Total Variance with Die and Coin)**.** Continuing Example 6.4: Y is a die roll, $X \mid (Y = y) \sim \text{Bin}(y, 1/2)$.

- $\mathbb{E}[X \mid Y] = Y/2$, so $\text{Var}(\mathbb{E}[X \mid Y]) = \text{Var}(Y/2) = \text{Var}(Y)/4 = (35/12)/4 = 35/48$.
- $\text{Var}(X \mid Y) = Y \cdot (1/2)(1/2) = Y/4$, so $\mathbb{E}[\text{Var}(X \mid Y)] = \mathbb{E}[Y]/4 = 3.5/4 = 7/8$.
- $\text{Var}(X) = 7/8 + 35/48 = 42/48 + 35/48 = 77/48 \approx 1.604$.

The within-group variance (7/8) and the between-group variance (35/48) together account for the total.

7 Exercises

Exercise 7.1 (Indicator Expectation and Probability)**.** $(*)$
Let $A_1, \ldots, A_n$ be events. Using the indicator random variables $\mathbf{1}_{A_i}$ and linearity of expectation, prove that

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} \mathbb{P}(A_i)$$

(the union bound). Compare this proof with the inclusion-exclusion approach from Lesson 9.

Exercise 7.2 (Variance of a Linear Combination). $(*)$
Let X_1, X_2, X_3 be random variables with $\mathrm{Var}(X_i) = \sigma^2$ for each i and $\mathrm{Cov}(X_i, X_j) = \rho\sigma^2$ for $i \neq j$, where $\rho \in [-1/2, 1]$.

(a) Compute $\mathrm{Var}(X_1 + X_2 + X_3)$ in terms of σ^2 and ρ.
(b) For what value of ρ is the variance of the sum minimized? What is the minimum?

Exercise 7.3 (Conditional Expectation Practice). $(**)$
A box contains N balls numbered $1, \ldots, N$, where N itself is random with $\mathbb{P}(N = n) = 1/4$ for $n = 1, 2, 3, 4$. A ball is drawn uniformly at random from the box; let X be its number.

(a) Compute $\mathbb{E}[X \mid N = n]$ for each n.
(b) Use the tower property to compute $\mathbb{E}[X]$.
(c) Use the law of total variance to compute $\mathrm{Var}(X)$.

Exercise 7.4 (Uncorrelated but Dependent – Discrete Case). $(**)$
Let Y be uniformly distributed on $\{-1, 0, 1\}$ and define $X = Y^2$.

(a) Compute $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\mathbb{E}[XY]$, and $\mathrm{Cov}(X, Y)$.
(b) Verify that X and Y are uncorrelated.
(c) Show that X and Y are *not* independent by finding values x, y such that $p_{X,Y}(x, y) \neq p_X(x)p_Y(y)$.

Exercise 7.5 (Tower Property and Bellman Equation). $(***)$
Consider an MDP with states $\{s_1, s_2\}$, actions $\{a\}$ (single action), discount factor $\gamma = 0.9$, reward $R(s_1) = 1$, $R(s_2) = 0$, and transition probabilities $\mathbb{P}(S_{t+1} = s_1 \mid S_t = s_1) = 0.8$ and $\mathbb{P}(S_{t+1} = s_1 \mid S_t = s_2) = 0.4$.

(a) Write out the Bellman equations for $V(s_1)$ and $V(s_2)$ using the tower property decomposition.
(b) Solve the system of linear equations for $V(s_1)$ and $V(s_2)$.
(c) Verify your answer by computing $V(s_1) = \mathbb{E}[R_0 \mid S_0 = s_1] + \gamma\,\mathbb{E}[V(S_1) \mid S_0 = s_1]$ explicitly.

Exercise 7.6 (Law of Total Variance for Mixture Distributions). $(***)$
A reward distribution is a mixture: with probability α, the reward is $R \sim \mathcal{N}(\mu_1, \sigma_1^2)$, and with probability $1 - \alpha$, the reward is $R \sim \mathcal{N}(\mu_2, \sigma_2^2)$. Let $Z \in \{1, 2\}$ indicate the component.

(a) Use the tower property to compute $\mathbb{E}[R]$.
(b) Use the law of total variance to compute $\mathrm{Var}(R)$ and show that it decomposes as

$$\mathrm{Var}(R) = \underbrace{\alpha\sigma_1^2 + (1-\alpha)\sigma_2^2}_{\text{within-component}} + \underbrace{\alpha(1-\alpha)(\mu_1 - \mu_2)^2}_{\text{between-component}}.$$

(c) Explain why mixture distributions always have higher variance than any single component when $\mu_1 \neq \mu_2$.

Hint: For part (b), compute $\mathrm{Var}(\mathbb{E}[R \mid Z])$ by noting that $\mathbb{E}[R \mid Z]$ takes two values with known probabilities.

8 Summary

- The *Law of the Unconscious Statistician* (LOTUS) states $\mathbb{E}[g(X)] = \sum_x g(x)p_X(x)$ (discrete) or $\mathbb{E}[g(X)] = \int g(x)f_X(x)\,dx$ (continuous), allowing computation of expectations of transformed variables without finding the new distribution.
- *Linearity of expectation* ($\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$) holds for *all* random variables with finite expectations, with *no independence assumption required.*
- Every probability can be expressed as the expectation of an indicator: $\mathbb{P}(A) = \mathbb{E}[\mathbf{1}_A]$.

- The *variance* satisfies $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$ and the computational formula $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.
- *Covariance* $\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$ is a symmetric, bilinear measure of linear association.
- The *variance of a sum* decomposes as $\mathrm{Var}(\sum X_i) = \sum \mathrm{Var}(X_i) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j)$; for independent variables, the cross-covariance terms vanish.
- Independence implies zero covariance, but *the converse is false*: uncorrelated variables can be strongly dependent (e.g., X and X^2 when X is symmetric around zero).
- The *conditional expectation* $\mathbb{E}[X \mid Y]$ is a random variable (a function of Y) that satisfies the *tower property*: $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$.
- The *law of total variance* $\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathbb{E}[X \mid Y])$ decomposes total variability into within-group and between-group components.

9 Further Reading

- **Lesson 12** develops the law of large numbers and the central limit theorem, which explain how sample means concentrate and why the normal distribution appears as a universal limit.
- **Phase 01, Lesson 5** (probability foundations) revisits expectation and conditional expectation in the measure-theoretic framework, defining $\mathbb{E}[X \mid \mathcal{G}]$ as a $\mathcal{G}$-measurable random variable satisfying a projection property. The tower property becomes a consequence of the σ-algebra nesting $\mathcal{G}_1 \subseteq \mathcal{G}_2$.
- **Phase 02** (Temporal Difference Learning, Policy Gradient Theorem) relies heavily on the linearity of expectation and the tower property for deriving TD update rules and the policy gradient theorem.
- For additional worked examples on conditional expectation and the tower property, see Blitzstein and Hwang [2019, Chapter 9], which includes the "conditional expectation as a projection" perspective.
- The law of total variance is sometimes called the *Eve's law* ("E of the V plus V of the E"); a thorough discussion with applications to hierarchical models appears in Bertsekas and Tsitsiklis [2008, Section 4.4].
- The Cauchy-Schwarz inequality for random variables ($|\mathbb{E}[XY]|^2 \le \mathbb{E}[X^2]\mathbb{E}[Y^2]$), which underlies the correlation bound $|\rho| \le 1$, has far-reaching consequences in functional analysis; see Phase 01 (Lesson 7, inner product spaces).

References

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